Climate-Smart Agriculture

Agriculture RAG Knowledge Base • Educational reference document

Climate-smart agriculture focuses on agricultural systems that can improve productivity and resilience while, where feasible, reducing greenhouse-gas emissions or increasing carbon storage. It is not a single technology; it is a framework for selecting context-appropriate practices.

Examples can include improved water management, soil-health practices, diversified cropping systems, stress-tolerant varieties, efficient nutrient management, agroforestry, conservation practices, and improved livestock management.

Climate risks differ across locations. Heat, drought, intense rainfall, flooding, salinity, and changing pest pressure can affect farms differently. Adaptation planning therefore begins with identifying the major local hazards and the sensitivity of the farming system.

Nutrient management is relevant to both productivity and environmental outcomes. Applying nutrients according to crop and soil needs can reduce waste and improve nutrient-use efficiency. Water management can improve resilience when irrigation or water harvesting is appropriate and sustainable.

Climate-smart agriculture should be evaluated using local evidence and trade-offs. A practice that works in one climate, soil type, or farming system may not produce the same result elsewhere. Farmers and advisors should consider cost, labor, water availability, expected benefits, risks, and local knowledge.

Reference basis

• FAO, Climate-Smart Agriculture resources

• FAO, irrigation and soil-management resources

Note: This document is a project knowledge-base document prepared for a student RAG demonstration. It is not a substitute for crop-specific agronomic advice, soil testing, product labels, or local agricultural recommendations.